

Mohammad Raditia Pradana

Researcher, Department of Geography,
Universitas Indonesia, Depok, Indonesia
M.Sc. in Geography Sciences

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Research Interests

Street-view imagery (SVI) • Urban data science • Computer vision • Geospatial analysis • Urban green spaces • Urban planning

Education

M.Sc. in Geography Sciences (Fast-Track Program) — Universitas Indonesia 2024–2025

GPA 4.00 / 4.00 | Summa Cum Laude | Ranked 1st in Cohort - Faculty level

Thesis: Spatial Modeling of Visual Element Proportion Diversity in Urban Spaces Based on Street-View and Spatial Indicators (Kebayoran Baru & Tanah Abang)

B.Sc. in Geography (Fast-Track Program) — Universitas Indonesia

2020–2024

GPA 3.85 / 4.00 | Cum Laude | Ranked 1st in Cohort - Department level

Thesis: Protection Level for Fire-Prone Areas in East Jakarta City

Publications

Journal Articles

- [1] Pradana, M.R. & Semedi, J.M. (2026). Indonesia's urban green spaces dilemma: Balancing law, ecology, and public space, highlighting Jakarta. *Human Geography*. [DOI]
- [2] Pradana, M.R., Dimyati, M. & Gamal, A. (2025). Harmonizing street-view semantics and spatial predictors for dominant urban visual composition modelling. *Computational Urban Science*, 5(5). [DOI]
- [3] Pradana, M.R., Wibowo, A. & Semedi, J.M. (2025). Multi-perspective evaluation of urban green views: Spatial and street-view data integration in Sudirman Central Business District, Indonesia. *Geomatics and Environmental Engineering*, 19(3), 91–114. [DOI]
- [4] Pradana, M.R., Dimyati, M. & Semedi, J.M. (2025). Unveiling greenery and visual comfort: Integrating Green View Index and image segmentation in panoramic rural landscapes. *GeoScape*, 19(2), 143–159. [DOI]
- [5] Pradana, M.R. & Dimyati, M. (2024). Tracking urban sprawl: A systematic review and bibliometric analysis of spatio-temporal patterns using remote sensing and GIS. *European Journal of Geography*, 15(3), 190–203. [DOI]
- [6] Gamal, A., Pradana, M.R., Wibisono, B.H., Navitas, P. & Aryal, J. (2025). Thematic fragmentation and convergence in urban flood simulation research: A 45-year bibliometric mapping. *Urban Science*, 9(7), 280. [DOI]

- [7] Semedi, J.M., **Pradana, M.R.**, Putera, D.A. & Rahatiningtyas, N.S. (2024). The impedance of East Jakarta road network to estimate fire station service areas. *Forum Geografi*, 38(2), 280–290. [\[DOI\]](#)

Conference Proceedings

- [1] **Pradana, M.R.** & Semedi, J.M. (2025). Effortless coastal monitoring: Unsupervised detection of shoreline alterations due to tin mining in Bangka Belitung. *IOP Conference Series: EES*, 1462(1). [\[DOI\]](#)
- [2] Ash-Shidiq, I.P., Supriatna, S., Darmawan, A., Warta, Z., Molidena, E., Valla, A., Hisan, Firdaus, M.I., Zakaria, N.A., Muafiroh, S., **Pradana, M.R.**, Putera, D.A. (2024). Biomass stock estimation using Landsat 8 imagery in Bukit Tigapuluh National Park, Riau. *IOP Conference Series: EES*, 1291(1). [\[DOI\]](#)

Incoming Publications

- [1] **Pradana, M.R.** (*In review*). Your Academic Career Depends on Your Co-Authors: When Single-Author Papers Do Not Count in Indonesia's Academic Promotion System. *Journal of Academic Ethics*.
- [2] **Pradana, M.R.**, Gamal. A., & Aryal. J. (*In review*). Reframing Urban "Sustainable" Streetscapes: Evidence from Cross-City Transformation and Convergence. *City and Environment Interactions*.
- [3] **Pradana, M.R.**, Gamal. A., & Aryal. J. (*In review*). Interpreting 'Sustainable' Urban Streetscapes with Generative AI: Context-Rich vs. Generic Prompting. *Urban Informatics*.

Experience

Researcher

May 2025 – Present

Department of Geography, Universitas Indonesia

- Co-PI on a university-funded grant (IDR 100 million) on climate resilience in urban parks via SVI and place attachment analysis.
- Mentors 3 undergraduate students across the full research cycle.

Full-time Researcher & Assistant Editor, *Smart City Journal*

Oct. 2024 – Jun. 2026

SMART CITY Universitas Indonesia — Research & Community Engagement Unit

- Geospatial research using SVI and generative AI for urban environment analysis.
- Manages *Smart City Journal* editorial workflow (manuscript handling, reviewer coordination).

Cartographer — SEA, West Bangka Regency

Mar. – Dec. 2025

Center for Social and Environmental Management Research (PPSML), Universitas Indonesia

- Solely responsible for producing 50+ thematic and base maps at multiple cartographic scales for the SEA document.

Part-time Research Assistant

Jun. – Oct. 2024

SMART CITY Universitas Indonesia — Research & Community Engagement Unit

- GIS and remote sensing analysis using 3D data and digital building audits.

- Project-in-Charge for a government-collaborated urban research project.

Cartographer — Detailed Spatial Planning, Baradatu District

May – Aug. 2024

Center for Applied Geography Research (PPGT), Universitas Indonesia

- Compiled 20+ large-scale base and thematic maps at 1:5,000 for the RDTR planning document.

Cartographer — SEA, Bangka Belitung Islands Province

Mar. – Oct. 2024

Center for Social and Environmental Management Research (PPSML), Universitas Indonesia

- Solely produced 40+ thematic maps including environmental sensitivity and infrastructure layers.

Student Researcher

Jun. 2023 – Oct. 2024

Department of Geography, Universitas Indonesia

- Fieldwork on urban fire management in Jakarta; produced 1 publication and 2 international conference presentations.

Cartographer — Regional Spatial Planning, Hulu Sungai Tengah Regency

Jan. – Apr. 2023

Center for Applied Geography Research (PPGT), Universitas Indonesia

- Produced 40+ thematic maps at 1:50,000 scale for the RTRW document, South Kalimantan.

Teaching

Master Programme of Urban Planning, Universitas Indonesia

Guest Lecturer — Data Management and Analysis

Apr. – May 2026

- Delivered technical lectures on database development, statistical analysis, and spatial analysis for graduate urban planning students.

Department of Geography, Universitas Indonesia

Teaching Assistant — GIS & Remote Sensing Practicals

Jun. – Aug. 2024

- Facilitated hands-on practicals for 30+ undergraduates across 6 self-developed modules covering core GIS and remote sensing workflows.

Research & Government Projects

Research Grants

Prompting Sustainability: A Comparative Analysis of How Generative AI Interprets and Visualizes Urban 'Sustainable' Streetscapes

Oct. 2025 – Present

Funded by Universitas Indonesia

Cooling Cities, Warming Communities: Quantifying Climate Resilience in Urban Parks Through Anthropomorphic Mapping and Place Attachment

May 2025 – Present

IDR 100 million, 2-year grant — Universitas Indonesia

Spatial Modeling of Visual Element Proportion Diversity [Master's Thesis]

Jun. 2024 – Jun. 2025

INSPIRASI Grant — SMART CITY UI

Collaborative Mapping & Mitigation for a Resilient Capital City with
Generative AI

Jun. 2024 – Dec. 2025

PRIME Engineering and Universitas Indonesia Grant

Level of Protection of Fire-Prone Areas in East Jakarta City

Aug. 2023 – Jan. 2024

Funded by Faculty of Mathematics and Natural Sciences, Universitas Indonesia

Government Projects

Innovative Government Awards 2025, Ministry of Home Affairs (*Lead data scientist*) Sep. – Nov. 2025

Strategic Environmental Assessment (SEA) — West Bangka Regency

Mar. – Dec. 2025

Innovative Government Awards 2024, Ministry of Home Affairs (*Lead data scientist*) Sep. – Oct. 2024

Strategic Environmental Assessment (SEA) — Bangka Belitung Islands Province

Mar. – Oct. 2024

Open-Source Tools & Datasets

Project Page

Prompting Sustainability — Research dissemination website

2025 - 2026 

Cooling Cities, Warming Communities — Research dissemination website

2025 - 2026 

Software, Tools, & Repositories

M2F Playground — All in one easy to access Notebook repo for Mask2Former inference

2026 

GSV Panorama Downloader — Notebook repo for batch downloading Google Street View panoramas

2025 

GSV Downloader by Links — Notebook repo for downloading Street View imagery via direct links

2025 

Facade Dominant Color Extractor — Notebook repo for extracting color information of building facade from street-view imagery

2025 

Open-Access Datasets

genai_sustainablestreetscape (v2) — Cross-city corpus of 4,186 paired images (raw + GPT-5-generated counterparts) from Jakarta (567), Melbourne (832), and Singapore (694).

2026 

Sustainable Streetscape Prompt-Specificity Dataset (v1) — 100 Jakarta GSV scenes × 3 treatments (raw, context-rich, non-context via GPT-4o), each paired with semantic segmentation masks.

2026 

Peer Review Service

<i>Discover Environment</i> , Springer Nature	June 2026
<i>Journal of Mountain Science</i> , Springer Nature	May - June 2026
<i>Scientific Reports</i> , Nature Publishing Group	Jan. – June. 2026
<i>Frontiers in Marine Sciences</i> , Frontiers	Jan. – Feb. 2026
<i>Urban Forestry & Urban Greening</i> , Elsevier	May – Aug. 2025
<i>Transactions in GIS</i> , Wiley	Jan. 2025

Conference Presentations

- Aug. 2024 The 2nd International Conference on Anthropocene, Global Environmental Change and Powerful Geography (Surakarta, Indonesia) — *Effortless Shoreline Detection in Tin-Mining Areas*. [Best Presenter Award]
- Nov. 2023 The 1st Aceh International Conference In Geography and Environmental Education (Aceh, Indonesia)— *The Impedance of East Jakarta Road Network to Estimate Fire Station Service Areas*.
- Nov. 2022 The 2nd International Conference of Science and Applied Geography (Depok, Indonesia) — *Biomass Stock Estimation Using Landsat 8 Imagery in Bukit Tigapuluh National Park*.

Honors & Awards

- 2025 **Ranked 1st in Cohort, M.Sc. Geography** — Summa Cum Laude (Faculty Level)
- 2024 **Ranked 1st in Cohort, B.Sc. Geography** — Cum Laude (Department Level)
- 2024 **Fast-Track Program Candidacy, B.Sc. - M.Sc. Geography** — selected based on academic excellence during undergraduate study
- 2024 **Best Presenter Award, ICoAGPG 2024**
- 2022 **2nd Place, National Economics Paper Competition (IMIE FEB UKSW)**

Technical Skills

Urban Analytics	Street-view imagery, object extraction through computer vision, urban data science, geospatial data science, spatial approach
Programming	Python (familiar), R (familiar)
Geospatial Platforms	ArcGIS Pro, QGIS, Google Earth Engine
Research Computing	Google Colab, Jupyter Notebook, VS Code, Git, GitHub, L ^A T _E X
Quantitative Methods	Statistical analysis, machine learning, computer vision, dimensionality reduction, systematic literature review, bibliometric analysis